

Probability PhD Exam, May 1989

1. Let X_n be independent identically distributed random variables and $S_n = \sum_{i=1}^n X_i$. Suppose

$$\mathbb{E} \left[\sup_n \left| \frac{S_n}{n} \right| \right] < \infty.$$

Show that

$$\mathbb{E}[|X_1| \log^+ |X_1|] < \infty,$$

where

$$\log^+ x = \begin{cases} \log x, & x \geq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Hint: $C_n = \prod_{j=1}^n \mathbb{P}[|X_j| \leq j] \rightarrow c > 0$. If $T = \inf\{n : |X_n| > n\}$, then

$$\sum_{n=1}^{\infty} \frac{1}{n} \int_{\{T=n\}} |X_n| dp = \sum_{n=1}^{\infty} \frac{C_{n-1}}{n} \int_{\{|X_1|>n\}} |X_1| dp$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n} \int_{\{T=n\}} |S_{n-1}| dp < \infty.$$

2. Let X_n be independent Poisson variables. Suppose $\sum_{n=1}^{\infty} X_n < \infty$ a.s. Show that $\sum_{n=1}^{\infty} \mathbb{E}[X_n] < \infty$.

Hint. Borel–Cantelli.

3. This problem has two parts.

- Define the notion of conditional expectation. In particular explain the notion $\mathbb{E}[X | Y]$ where X and Y are random variables.
- Suppose X, Y are random variables such that $\mathbb{E}(X^2), \mathbb{E}(Y^2) < \infty$ and

$$\mathbb{E}[X | Y] = Y, \quad \mathbb{E}[Y | X] = X.$$

Show that $X = Y$ a.s.

4. **Warning: Suspected error.** Part (b) is retained as printed, but its conclusion can fail for a nonconstant symmetric characteristic function when the sequence alternates in sign. Any intended additional restriction or alternative conclusion is not uniquely determined.

This problem has three parts.

- A sequence of characteristic functions on $\mathbb{R}^1$ converging pointwise to a characteristic function does so uniformly on compact sets. Explain why - no need to give complete proofs.
- Let $b_n \in \mathbb{R}$ and f a characteristic function not identically equal to 1. Suppose $f(b_n t)$ converges to a characteristic function. Show that $\lim b_n = b \in \mathbb{R}$.
- Let $\phi(t) = \frac{1}{8}[1 + 7e^{it}]$. Show that ϕ is a characteristic function but that $|\phi|$ is not.

Hint. $|\phi|^2$ is the characteristic function of a measure concentrated on $\{-1, 0, 1\}$. If $|\phi|$ is the characteristic function of a measure m , then $|\phi|^2$ is the characteristic function of $m * m$.

5. Let X_t be a standard Brownian motion.

- (i) Show that (X_t) , $(X_t^2 - t)$, and $e^{uX_t - \frac{1}{2}u^2t}$ are all martingales.
- (ii) Let $T_a(w) = \inf\{t : X_t(w) = a\}$ for $a > 0$, the inf being defined to be $+\infty$ if the set is empty. Using (i), compute $E[e^{-\lambda T_a}]$ for $\lambda > 0$, and show that $T_a < \infty$ a.s. and $E[T_a] = \infty$.

6. State and sketch the proof of the martingale convergence theorem.

7. Using the martingale convergence theorem prove the following:

Let F_n be an increasing sequence of σ -fields, X_n a sequence of random variables dominated by an integrable random variable Y : $|X_n| \leq Y$. Suppose $X_n \rightarrow X$ a.s. Then

$$E[X_n | F_n] \rightarrow E[X | F_\infty] \quad \text{a.s.}$$

Hint. Put the

$$U_n = \sup_{m \geq n} |X_m - X|.$$

Then for $m \geq n$,

$$E[|X_m - X| | F_m] \leq E[U_n | F_m].$$

$$E \left\{ \limsup_m E[|X_m - X| | F_m] \right\} \leq E\{E[U_n | F_\infty]\} = E[U_n].$$